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Semiconductor package system

Abstract

Provided is a semiconductor package system. The system includes a substrate, a first semiconductor package on the substrate, a second semiconductor package on the substrate, a first passive element on the substrate, a heat dissipation structure on the first semiconductor package, the second semiconductor package, and the first passive element, and a first heat conduction layer between the first semiconductor package and the heat dissipation structure. A sum of a height of the first semiconductor package and a thickness of the first heat conduction layer may be greater than a height of the first passive element. The height of the first semiconductor package may be greater than a height of the second semiconductor package.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This U.S. non-provisional patent application is a continuation of U.S. application Ser. No. 16/397,278, filed Apr. 29, 2019, which claims priority under 35 U.S.C. § 119 to Korean Patent Application Nos. 10-2018-0054304, filed on May 11, 2018; 10-2018-0054305, filed on May 11, 2018; 10-2018-0054307, filed on May 11, 2018; 10-2018-0055081, filed on May 14, 2018; and 10-2018-0110511, filed on Sep. 14, 2018, the entire contents of each of which are hereby incorporated by reference.

BACKGROUND
(1) The present disclosure herein relates to a semiconductor package system, and more particularly to a semiconductor package system having a heat dissipation structure.
(2) The semiconductor package is implemented in a form suitable for use in an electronic product. Generally, semiconductor packages are generally mounted with a semiconductor chip on a printed circuit board (PCB) and electrically connected to each other using bonding wires or bumps. As the semiconductor package is increased in speed and capacity, the power consumption of the semiconductor package is increasing. Accordingly, the thermal characteristics of the semiconductor package become more important.
SUMMARY
(3) Inventive concepts relate to a semiconductor package having improved thermal characteristics and a semiconductor module including the same.
(4) According to an embodiment of inventive concepts, a semiconductor package system may include a substrate; a first semiconductor package on the substrate; a second semiconductor package on the substrate; a first passive element on the substrate; a heat dissipation structure provided on the first semiconductor package, the second semiconductor package, and the first passive element; and a first heat conduction layer between the first semiconductor package and the heat dissipation structure. A sum of a height of the first semiconductor package and a thickness of the first heat conduction layer may be greater than a height of the first passive element. The height of the first semiconductor package may be greater than a height of the second semiconductor package.
(5) In an embodiment of inventive concepts, a semiconductor package system may include a substrate; a first semiconductor package on an upper surface of the substrate, and the first semiconductor package including a first semiconductor chip, the first semiconductor chip including one or more logic circuits; a second semiconductor package on the upper surface of the substrate; a passive element on the upper surface of the substrate; a heat dissipation structure on the first semiconductor package, the second semiconductor package, and the passive element; and a plurality of heat conduction layers that are each in physical contact with a lower surface of the heat

dissipation structure. The plurality of heat conduction layers may include a first heat conduction layer on an upper surface of the first semiconductor package, and the first heat conduction layer may have a thinnest thickness among the plurality of heat conduction layers.
(6) In an embodiment of inventive concepts, a semiconductor package system may include a substrate; a first semiconductor package on the substrate, the first semiconductor package including a first semiconductor chip, the first semiconductor chip including one or more logic circuits; a second semiconductor package on the substrate; a passive element on the substrate; a heat dissipation structure on the first semiconductor package, the second semiconductor package, and the passive element; a first heat conduction layer on the first semiconductor package, the first heat conduction layer in physical contact with the heat dissipation structure; and a second heat conduction layer on the second semiconductor package, the second heat conduction layer in physical contact with the heat dissipation structure. A thickness of the first heat conduction layer may be a smaller than a thickness of the second heat conduction layer. An upper surface of the first heat conduction layer may be provided at a level higher than an upper surface of the passive element.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The accompanying drawings are included to provide a further understanding of inventive concepts, and are incorporated in and constitute a part of this specification. The drawings illustrate example embodiments of inventive concepts and, together with the description, serve to explain principles of inventive concepts. In the drawings:
(2) FIG. **1**A is a plan view showing a package system according to example embodiments;
(3) FIG. **1**B is a plan view showing a package system according to example embodiments;
(4) FIG. **1**C is a cross-sectional view taken along the line I-II of FIG. **1**A;
(5) FIG. **1**D is an enlarged view of the region A of FIG. **1**C;
(6) FIG. **1**E is an enlarged view of the region B of FIG. **1**C;
(7) FIG. **1**F is a view showing a package system according to example embodiments;
(8) FIG. **1**G corresponds to the enlarged view of the region III of FIG. **1**A;
(9) FIG. **1**H is a cross-sectional view taken along the line I′-II′ of FIG. **1**G;
(10) FIG. **1**I is a view for explaining a first semiconductor package according to example embodiments;
(11) FIG. **2**A is a plan view showing a package system according to example embodiments;
(12) FIG. **2**B is a cross-sectional view taken along the line I-II of FIG. **2**A;
(13) FIG. **2**C is a plan view showing a package system according to example embodiments;
(14) FIG. **2**D is a cross-sectional view taken along the line I-II of FIG. **2**C;
(15) FIG. **2**E is a cross-sectional view showing a package system according to example embodiments;
(16) FIG. **3**A is a cross-sectional view showing a package system according to example embodiments;
(17) FIG. **3**B is a cross-sectional view showing a package system according to example embodiments;
(18) FIG. **3**C is a cross-sectional view showing a package system according to example embodiments;
(19) FIG. **4**A is a cross-sectional view showing a package system according to example embodiments;
(20) FIG. **4**B is a cross-sectional view showing a package system according to example embodiments;

(21) FIG. **4**C is a cross-sectional view showing a package system according to example embodiments;
(22) FIG. **5**A is a cross-sectional view showing a semiconductor module according to example embodiments;
(23) FIG. **5**B is a view for explaining a second passive element according to example embodiments, and is a cross-sectional view showing an enlarged view of the region C of FIG. **5**A;
(24) FIG. **5**C is a view for explaining lower pads and conductive terminals according to example embodiments; and
(25) FIG. **5**D is a view for explaining lower pads according to example embodiments.
DETAILED DESCRIPTION
(26) In this specification, like reference numerals refer to like components throughout the specification. Hereinafter, a package system according to inventive concepts and a semiconductor module including the same will be described. The semiconductor package system may be a package system or a semiconductor module including the package system.
(27) FIG. **1**A is a plan view showing a package system according to example embodiments. FIG. **1**B is a plan view showing a package system according to example embodiments. FIG. **1**C is a cross-sectional view taken along the line I-II of FIG. **1**A. FIG. **1**D is an enlarged view of the region A of FIG. **1**C. FIG. **1**E is an enlarged view of the region B of FIG. **1**C.
(28) Referring to FIGS. **1**A, **1**B, **1**C, **1**D, and **1**E, a package system **1** includes a substrate **500**, a first semiconductor package **100**, a second semiconductor package **200**, a third semiconductor package **300**, a first passive element **400**, a heat dissipation structure **600**, and a first heat conduction layer **710**. As an example, a printed circuit board (PCB) having a circuit pattern may be used as a substrate **500**. Conductive terminals **550** may be provided on the lower surface of the substrate **500**. The conductive terminals **550** may include at least one of solder balls, bumps, and pillars. The conductive terminals **550** may include, for example, a metal.
(29) The first semiconductor package **100** may be mounted on the upper surface **500***a* of the substrate **500**. The first semiconductor package **100** may include a logic chip or a system-on-chip, as described later. The first connection terminals **150** may be interposed between the substrate **500** and the first semiconductor package **100**. The first semiconductor package **100** may be electrically connected to the substrate **500** through the first connection terminals **150**. In this specification, the electrical connection with the substrate **500** may mean that it is electrically connected with the interconnections **505** in the substrate **500**. The first connection terminals **150** may include a solder ball, a pillar, a bump, or a ball grid array. The height H**1** of the mounted first semiconductor package **100** may be defined as including the height of the first connection terminals **150**. In this specification, the height of any component may mean the maximum distance of the component measured in a direction perpendicular to the upper surface **500***a* of the substrate **500**. The pitch of the first connection terminals **150** may be smaller than the pitch of the conductive terminals **550**.
(30) The second semiconductor package **200** may be mounted on the upper surface **500***a* of the substrate **500**. The second semiconductor package **200** may be spaced apart from the first semiconductor package **100** in plan view. The second semiconductor package **200** may be a semiconductor package different type from the first semiconductor package **100**. The second connection terminals **250** may be interposed between the substrate **500** and the second semiconductor package **200**. The second semiconductor package **200** may be electrically connected to the substrate **500** through the second connection terminals **250**. The second connection terminals **250** may include a solder ball, a pillar, a bump, or a ball grid array. The pitch of the second connection terminals **250** may be smaller than the pitch of the conductive terminals **550**. The height H**2** of the mounted second semiconductor package **200** may be defined as including the height of the second connection terminals **250**. The second semiconductor package **200** may be provided in plurality. The second semiconductor packages **200** may be spaced apart from each other. However, the number and the planar arrangement of the second semiconductor packages **200** may be

variously modified.

(31) The third semiconductor package **300** may be mounted on the substrate **500**. The third semiconductor package **300** may be spaced apart from the first semiconductor package **100** and the second semiconductor package **200** in plan view. The third semiconductor package **300** may be a semiconductor package different type from the first and second semiconductor packages **100** and **200**. The third semiconductor package **300** may be provided in single as shown in FIG. **1**A. As another example, the third semiconductor package **300** may be provided in plurality as shown in FIG. **1**B. In this case, the third semiconductor packages **300** may be spaced apart from each other. The number and planar arrangement of the third semiconductor packages **300** may be variously modified without being limited to those shown in FIGS. **1**A and **1**B. Hereinafter, the third semiconductor package **300** in single will be described. The third connection terminals **350** may be interposed between the substrate **500** and the third semiconductor package **300** as shown in FIG. **1**C. The third semiconductor package **300** may be electrically connected to the substrate **500** through the third connection terminals **350**. The third connection terminals **350** may include a solder ball, a pillar, a bump, or a ball grid array. The pitch of the third connection terminals **350** may be smaller than the pitch of the conductive terminals **550**. The height H**3** of the mounted third semiconductor package **300** may be defined as including the height of the third connection terminals **350**. The height H**1** of the mounted first semiconductor package **100** may be greater than the height H**3** of the mounted third semiconductor package **300**.

(32) The first passive element **400** may be mounted on the upper surface **500***a* of the substrate **500**. The first passive element **400** may be spaced apart from the first, second, and third semiconductor packages **100**, **200**, and **300** in plan view. The first passive element **400** may include any one of an inductor, a resistor, and a capacitor. The first connection terminal portions **401** may be further provided between the substrate **500** and the first passive element **400** as shown in FIG. **1**D. The first connection terminal portions **401** may include, for example, a solder, a pillar, a bump, or a ball grid array. The height H**4** of the mounted first passive element **400** may be defined as including the height of the first connection terminal portions **401**. For example, the height H**4** of the first passive element **400** may be equal to the sum of the height H**41** of the first connection terminal portions **401** and the height H**40** of the first passive element **400**′ before being mounted. The height H**4** of the mounted first passive element **400** may be substantially equal to the distance between the upper surface **500***a* of the substrate **500** and the uppermost surface of the first passive element **400**. The first passive element **400** may be provided in plurality. As shown in FIGS. **1**A and **1**B, the first passive elements **400** may be spaced apart from each other. The number and the planar arrangement of the first passive elements **400** may be variously modified. Hereinafter, the single first passive element **400** will be described. In the drawings other than FIG. **1**D, the first connection terminal portions **401** are omitted for simplification, but inventive concepts are not limited thereto.

(33) A heat dissipation structure **600** may be provided on the first to third semiconductor packages **100**, **200**, **300** and the first passive element **400**. The lower surface **600***b* of the heat dissipation structure **600** may face the first, second, and third semiconductor packages **100**, **200**, and **300**. The lower surface **600***b* of the heat dissipation structure **600** may be substantially flat. For example, the lower surface **600***b* of the heat dissipation structure **600** on the first semiconductor package **100**, the lower surface **600***b* of the heat dissipation structure **600** above on the second semiconductor package **200**, the lower surface **600***b* on the third semiconductor package **300**, and the lower surface **600***b* of the heat dissipation structure **600** above on the first passive element **400** may be disposed at substantially the same level. The additional processing on the lower surface **600***b* of the heat dissipation structure **600** is omitted, so that the manufacture of the heat dissipation structure **600** may be simplified. The processing may include forming a trench or forming a protrusion. The heat dissipation structure **600** may include a thermally conductive material. The thermally conductive material may include a metal (e.g., copper and/or aluminum) or a carbon containing material (e.g., graphene, graphite, and/or carbon nanotubes). The heat dissipation structure **600** may

have a relatively high thermal conductivity. As an example, a single metal layer or a plurality of stacked metal layers may be used as the heat dissipation structure **600**. As another example, the heat dissipation structure **600** may include a heat sink or a heatpipe. As another example, the heat dissipation structure **600** may use a water cooling method. The heat dissipation structure **600** may include a first heat dissipation structure **610**. The first heat dissipation structure **610** may be spaced apart from the substrate **500**.

(34) The first heat conduction layer **710** may be interposed between the first semiconductor package **100** and the heat dissipation structure **600**. The first heat conduction layer **710** may be in physical contact with the upper surface of the first semiconductor package **100** and the lower surface **600***b* of the heat dissipation structure **600**. The first heat conduction layer **710** may include a thermal interface material (TIM). The thermal interface material may include, for example, polymers and thermally conductive particles. The thermally conductive particles may be dispersed within the polymer. During an operation of the first semiconductor package **100**, heat generated from the first semiconductor package **100** may be transferred to the heat dissipation structure **600** through the first heat conduction layer **710**.

(35) According to example embodiments, the sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710** may be greater than the height H**4** of the mounted first passive element **400**. Even if the first passive element **400** is provided on the upper surface **500***a* of the substrate **500**, the first heat conduction layer **710** may be in physical contact with the first semiconductor package **100** and the heat dissipation structure **600**.

(36) The second heat conduction layer **720** may be provided between the second semiconductor package **200** and the heat dissipation structure **600**. The second heat conduction layer **720** may be in physical contact with the upper surface of the second semiconductor package **200** and the lower surface **600***b* of the heat dissipation structure **600**. The second heat conduction layer **720**, for example, may include a thermal interface material. During an operation of the second semiconductor package **200**, heat generated from the second semiconductor package **200** may be transferred to the heat dissipation structure **600** through the second heat conduction layer **720**.

(37) The third heat conduction layer **730** may be provided between the third semiconductor package **300** and the heat dissipation structure **600**. The third heat conduction layer **730** may be in physical contact with the upper surface of the third semiconductor package **300** and the lower surface **600***b* of the heat dissipation structure **600**. The third heat conduction layer **730**, for example, may include a thermal interface material. During an operation of the third semiconductor package **300**, heat generated from the third semiconductor package **300** may be transferred to the third semiconductor package **300** through the third heat conduction layer **730**.

(38) During an operation of the package system **1**, a lot of heat may be generated from the first semiconductor package **100**. For example, the first semiconductor package **100** may generate more heat than those from the second semiconductor package **200**, the third semiconductor package **300**, and the first passive element **400**. The thermal characteristics of the first semiconductor package **100** may have a greater effect on the operating characteristics of the package system **1** than the thermal characteristics of the second and third semiconductor packages **200** and **300**. As the thermal characteristics of the first semiconductor package **100** are improved, the operating characteristics of the package system **1** may be improved. Each of the first to third heat conduction layers **710**, **720**, **730** may have a lower thermal conductivity than that of the heat dissipation structure **600**. As the thickness A**1** of the first heat conduction layer **710** decreases, the heat generated from the first semiconductor package **100** may be emitted more quickly to the heat dissipation structure **600**. According to example embodiments, the thickness A**1** of the first heat conduction layer **710** may be the smallest among the thicknesses of the heat conduction layers contacting the lower surface **600***b* of the heat dissipation structure **600**. Here, the heat conduction layers may include first to third heat conduction layers **710**, **720**, and **730**. The heat conduction

layers may further include conductive adhesive patterns **741**, which will be described later with reference to FIGS. **2**A to **2**D. The thickness A**1** of the first heat conduction layer **710** may be less than the thickness A**2** of the second heat conduction layer **720** and the thickness A**3** of the third heat conduction layer **730**, for example. Accordingly, the heat generated from the first semiconductor package **100** may be transferred to the heat dissipation structure **600** more quickly. The package system **1** may show improved operating characteristics.

(39) An electronic element **430** may further be provided on the upper surface **500***a* of the substrate **500**. The electronic element **430** may include an oscillator such as a crystal oscillator or a real-time clock. As shown in FIG. **1**E, a conductive connection terminal **403** may be further provided between the electronic element **430** and the upper surface **500***a* of the substrate **500** so as to be electrically connected to the electronic element **430** and the substrate **500**. The height H**5** of the mounted electronic element **430** may be defined as including the height H**51** of the conductive connection terminal **403**. The height H**5** of the mounted electronic element **430** may be equal to the sum of the height H**51** of the conductive connection terminal **403** and the height H**50** of the electronic element **430**′ before being mounted. The sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710** may be greater than the height H**5** of the mounted electronic element **430**. Although the electronic element **430** is provided on the upper surface **500***a* of the substrate **500**, the heat generated from the first semiconductor package **100** may be smoothly discharged to the heat dissipation structure **600** through the first heat conduction layer **710**. As another example, the electronic element **430** may not be provided. In the drawings other than FIG. **1**E, the conductive connection terminal **403** is omitted for simplification, but inventive concepts are not limited thereto. Hereinafter, the electrical connection of the semiconductor packages **100**, **200**, and **300** will be described.

(40) The first semiconductor package **100** is electrically connected to the second semiconductor package **200**, the third semiconductor package **300** and the conductive terminals **550** through the interconnections **505** of the substrate **500** as shown in FIG. **1**C. The second semiconductor package **200** may be electrically connected to the first semiconductor package **100**, the third semiconductor package **300**, and the conductive terminals **550** through the interconnections **505** of the substrate **500**. The third semiconductor package **300** may be electrically connected to the first semiconductor package **100**, the second semiconductor package **200**, and the conductive terminals **550** through the interconnections **505** of the substrate **500**.

(41) The first underfill film **160** may be provided in the gap between the substrate **500** and the first semiconductor package **100** to seal the first connection terminals **150**. A second underfill film **260** may be provided in the gap between the substrate **500** and the second semiconductor package **200** to seal the second connection terminals **250**. A third underfill film **360** may be provided in the gap between the substrate **500** and the second semiconductor package **200** to seal the third connection terminals **350**. The first to third underfill films **160**, **260**, and **360** may include an insulating polymer such as an epoxy-based polymer. As the first to third underfill films **160**, **260**, and **360** are provided, the reliability of bonding of the first to third connection terminals **150**, **250**, and **350** may be improved. Unlike the illustrated embodiment, at least one of the first to third underfill films **160**, **260**, and **360** may be omitted.

(42) A dam structure **590** may be further provided on the upper surface **500***a* of the substrate **500**. The dam structure **590** may be disposed between the third semiconductor package **300** and the first passive element **400**. The dam structure **590** may be formed by using a liquid resin. Although not shown in the drawing, the substrate **500** may include a plurality of layers, and the uppermost layer of the layers may include an insulating polymer such as a solder resist material. In one example, the dam structure **590** may be formed integrally with the uppermost layer of the substrate **500**. In this case, the dam structure **590** may be connected to the uppermost layer of the substrate **500** without an interface. As another example, the dam structure **590** may include a material different form that of substrate **500**. For example, the dam structure **590** may be formed of the same material as any

one of the first to third underfill films **160**, **260**, and **360**. The height of the dam structure **590** may be equal to or less than the sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710**.
(43) The arrangement and number of the dam structures **590** may be variously modified. For example, the dam structure **590** may be disposed between the first semiconductor package **100** and the first passive element **400**. As another example, the dam structure **590** may be disposed between the second semiconductor package **200** and the first passive element **400**. The dam structure **590** may be provided in plurality as shown in FIG. **1**A. The dam structures **590** may be spaced apart from one another. Hereinafter, each of the first to third semiconductor packages **100**, **200**, and **300** will be described in more detail.
(44) FIG. **1**F is a view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **1**A. Hereinafter, the contents overlapping with those described above will be omitted. In the description of FIG. **1**F, FIGS. **1**A, **1**B, and **1**C are described together.
(45) Referring to FIG. **1**F, a package system **1***a* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**.
(46) The first semiconductor package **100** may include a first substrate **110**, a first semiconductor chip **120**, and a first molding layer **130**. As an example, a printed circuit board (PCB) may be used as the substrate **500**. As another example, a redistribution layer may be used as the substrate **500**. The first semiconductor chip **120** may be flip-chip mounted on the first substrate **110**. Connection portions may be provided between the first semiconductor chip **120** and the first substrate **110**. The connection portions may include a solder ball, a pillar, a bump, or a ball grid array. The first semiconductor chip **120** may be a system on chip (SOC), a logic chip, or an application processor (AP) chip. The first semiconductor chip **120** may include circuits having different functions. The first semiconductor chip **120** may include a logic circuit and a memory circuit. The first semiconductor chip **120** may further include at least one of a digital integrated circuit (IC), a wireless radio frequency integrated circuit (RFIC), and an input/output circuit. Generating heat from the first semiconductor package **100** may mean that heat is generated from the first semiconductor chip **120**.
(47) The first molding layer **130** may be disposed on the first substrate **110** to cover the first semiconductor chip **120**. The first molding layer **130** covers the side surfaces and the upper surface of the first semiconductor chip **120** to seal the first semiconductor chip **120**. In this case, the upper surface of the first semiconductor package **100** may correspond to the upper surface of the first molding layer **130**. The first molding layer **130** may include an insulating polymer such as an epoxy molding compound. The first molding layer **130** may further extend into the gap between the first substrate **110** and the first semiconductor chip **120**. Unlike what is shown, an additional underfill pattern may be filled in the gap between the first substrate **110** and the first semiconductor chip **120**. The underfill pattern may be formed by a method of thermal compression of a nonconductive paste or a nonconductive film or by a capillary underfill process. The height H**1** of the mounted first semiconductor package **100** is defined as the sum of the height of the first connection terminals **150**, the height of the first substrate **110**, and the height of the first molding layer **130**.
(48) The second semiconductor package **200** may include a second substrate **210**, a second semiconductor chip **220**, and a second molding layer **230**. A printed circuit board (PCB) or redistribution layer may be used as the substrate **500**. The second semiconductor chip **220** may be a semiconductor chip different type from the first semiconductor chip **120**. For example, the second semiconductor chip **220** may function as a memory chip. The memory chip may include a DRAM chip. As another example, the memory chip may include SRAM, MRAM, and/or NAND flash memory. Generating heat from the second semiconductor package **200** may mean that heat is

generated from the second semiconductor chip **220**. The second semiconductor chip **220** may be mounted by a flip chip method or a bonding wire method. When the second semiconductor chip **220** is flip-chip mounted, an additional underfill pattern may be filled in the gap between the second substrate **210** and the second semiconductor chip **220**. The second semiconductor package **200** may include a plurality of second semiconductor chips **220**. As another example, the second semiconductor package **200** may include a single second semiconductor chip **220**. The second molding layer **230** covers the side surfaces and the upper surface of the second semiconductor chip **220** to seal the second semiconductor chip **220**. In this case, the upper surface of the second semiconductor package **200** may correspond to the upper surface of the second molding layer **230**. Unlike what is shown, the second molding layer **230** covers the side surface of the second semiconductor chip **220**, and may expose the upper surface of the second semiconductor chip **220**. In this case, the upper surface of the second semiconductor package **200** may correspond to the upper surface of the second molding layer **230** and the upper surface of the second semiconductor chip **220** exposed by the second molding layer **230**. The second molding layer **230** may include an insulating polymer such as an epoxy-based polymer. The height H**2** of the mounted second semiconductor package **200** is defined as the sum of the height of the second connection terminals **250**, the height of the second substrate **210**, and the height of the second molding layer **230**.
(49) The third semiconductor package **300** may include a third substrate **310**, a third semiconductor chip **320**, and a third molding layer **330**. A redistribution layer or a printed circuit board may be used as the third substrate **310**. When a redistribution layer is used as the third substrate **310**, the third semiconductor package **300** may be fabricated with a fan-out panel level package or a fan-out wafer level package. The third semiconductor chip **320** may be a semiconductor chip different type from the first semiconductor chip **120** and the second semiconductor chip **220**. For example, the third semiconductor chip **320** may include a power management integrated circuit (PMIC) to function as a power management chip. Generating heat from the third semiconductor package **300** may mean that heat is generated from the third semiconductor chip **320**. The third molding layer **330** may be provided on the third substrate **310** to cover the upper surface and side surfaces of the third semiconductor chip **320**. In this case, the upper surface of the third semiconductor package **300** may correspond to the upper surface of the third molding layer **330**. Unlike what is shown, the third molding layer **330** covers the side surface of the third semiconductor chip **320**, and may expose the upper surface of the third semiconductor chip **320**. In this case, the upper surface of the third semiconductor package **300** may correspond to the upper surface of the third molding layer **330** and the upper surface of the third semiconductor chip **320** exposed by the third molding layer **330**. The third molding layer **330** may include an insulating polymer such as an epoxy-based polymer. The height H**3** of the mounted third semiconductor package **300** is defined as the sum of the height of the third connection terminals **350**, the height of the third substrate **310**, and the height of the third molding layer **330**. The formation of the third semiconductor package **300** may include providing the third semiconductor chip **320** on a carrier substrate, forming the third molding layer **330** covering the third semiconductor chip **320**, removing the carrier substrate to expose a lower surface of the third semiconductor chip **320**, and forming a redistribution layer on the lower surface of the exposed third semiconductor chip **320** and a lower surface of the molding layer. In this case, the redistribution layer may be a third substrate **310**.
(50) FIG. **1**G corresponds to the enlarged view of the region III of FIG. **1**A. FIG. **1**H is a cross-sectional view taken along the line I′-II′ of FIG. **1**G. In the following description, FIGS. **1**A, **1**B, **1**C, and **1**D are referenced together.
(51) Referring to FIGS. **1**G and **1**H, a first marker **139** may be provided on the first molding layer **130**. For example, the first marker **139** may be provided on the upper surface of the first molding layer **130**. Unlike this, the first marker **139** may be provided on the side surface of the first molding layer **130**. The first marker **139** may be a recessed portion on one surface of the first molding layer **130**. Formation of the first marker **139** may include removing a portion of the first molding layer

130. When the first marker **139** is formed on the first semiconductor chip **120**, the first semiconductor chip **120** may be damaged during the formation of the first marker **139**. For example, a crack may be formed on the first semiconductor chip **120** or in the first semiconductor chip **120**. According to example embodiments, the first marker **139** may be provided on the first molding layer **130**, so that the first semiconductor chip **120** may not be damaged in the process of forming the first marker **139**. The first marker **139** may provide and display information about the first semiconductor package **100**. In the drawings other than FIG. **1**G to FIG. **1**I, the first marker **139** is omitted for convenience, but inventive concepts are not limited thereto.
(52) A first heat conduction layer **710** may be formed on the upper surface of the first semiconductor package **100**. Formation of the first heat conduction layer **710** may include providing a thermal interface material on the first semiconductor package **100** and then curing the thermal interface material. The thermal interface material prior to curing may have fluidity. In the process of forming the first heat conduction layer **710**, even if the thermal interface material on the edge region of the upper surface of the first semiconductor package **100** flows down to the side surface **100***c* of the first semiconductor package **100**, the thermal interface material on the center region of the upper surface of the first semiconductor package **100** may not flow down. Thus, the first heat conduction layer **710** may well fill the gap between the center region of the upper surface of the first semiconductor package **100** and the heat dissipation structure **600**. For example, the upper surface **710***a* of the first heat conduction layer **710** in the center region of the first semiconductor package **100** may be in physical contact with the heat dissipation structure **600**. According to example embodiments, since the first molding layer **130** is provided, the first semiconductor chip **120** may be provided in the center region of the first semiconductor package **100** in plan view. Accordingly, even if the thermal interface material partly flows down in the process of forming the first heat conduction layer **710**, the first heat conduction layer **710** may well transfer the heat of the first semiconductor chip **120** to the first heat dissipation structure **610**. When the first molding layer **130** includes the first marker **139**, the first heat conduction layer **710** may extend into the first marker **139**. Referring to FIG. **1**C, a second heat conduction layer **720** may be provided on the upper surface of the second molding layer **230**. The formation of the second heat conduction layer **720** may be performed through substantially the same method described in the formation of the first heat conduction layer **710**. Although the thermal interface material partly flows down during the formation of the second heat conduction layer **720**, the second heat conduction layer **720** may well fill the gap between the center region of the upper surface of the second semiconductor package **200** and the heat dissipation structure **600**. The center region of the second semiconductor package **200** may be a region provided with the second semiconductor chip **220**. Accordingly, the heat generated from the second semiconductor chip **220** may be well emitted to the heat dissipation structure **600** through the second heat conduction layer **720**.
(53) Although not shown in the drawing, a second marker may be further provided on the second molding layer **230**. The second marker may be the recessed portion of the second molding layer **230**.
(54) A third heat conduction layer **730** may be formed on the upper surface of the third molding layer **330**. The formation of the third heat conduction layer **730** may be performed through substantially the same method described in the formation of the first heat conduction layer **710**. At this time, although the thermal interface material partly flows down during the formation of the third heat conduction layer **730**, the third heat conduction layer **730** may well fill the gap between the center region of the upper surface of the third semiconductor package **300** and the heat dissipation structure **600**. The center region of the third semiconductor package **300** may be a region provided with the third semiconductor chip **320**. Accordingly, the thermal characteristics of the third semiconductor package **300** may be improved. Although not shown in the drawing, a third marker may be further provided on the third molding layer **330**. The third marker may be the

recessed portion of the third molding layer **330**.

(55) FIG. **1**I is a view for explaining a first semiconductor package according to example embodiments, and corresponds to a cross-section taken alone the line I′-II′ of FIG. **1**G.

(56) Referring to FIGS. **1**G and **1**I, the first semiconductor package **100** may include a first substrate **110**, a first semiconductor chip **120**, and a first molding layer **130**. The first molding layer **130** covers the side surface of the first semiconductor chip **120**, and may expose the upper surface of the first semiconductor chip **120**. In this case, the upper surface of the first semiconductor package **100** may correspond to the upper surface of the first molding layer **130** and the upper surface of the first semiconductor chip **120** exposed by the first molding layer **130**. The upper surface of the exposed first semiconductor chip **120** may be in direct physical contact with the first heat conduction layer **710**. Heat generated from the first semiconductor chip **120** may be transferred to the heat dissipation structure **600** through the first heat conduction layer **710**. Accordingly, the heat dissipation characteristic of the first semiconductor chip **120** may be further improved.

(57) FIG. **2**A is a plan view showing a package system according to example embodiments. FIG. **2**B is a cross-sectional view taken along the line I-II of FIG. **2**A. Hereinafter, the contents overlapping with those described above will be omitted.

(58) Referring to FIGS. **2**A and **2**B, a package system **1***b* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The substrate **500**, the first to third semiconductor packages **100**, **200** and **300**, the first passive element **400** and the first to third heat conduction layers **710**, **720**, and **730** may be substantially the same as those described above with reference to FIGS. **1**A to **1**I.

(59) A ground pattern may be provided on the upper surface **500***a* of the substrate **500**. The ground pattern may include a ground pad **510**G. At least one of the conductive terminals **550** may function as a ground terminal. A ground voltage may be applied to the ground pad **510**G through the ground terminal and the substrate **500**.

(60) The heat dissipation structure **600** may include a second heat dissipation structure **620**. The second heat dissipation structure **620** may include a body portion **621** and a leg portion **622**. The body portion **621** of the second heat dissipation structure **620** may be similar to the first heat dissipation structure **610** previously described with reference to FIGS. **1**A to **1**C. The lower surface **600***b* of the heat dissipation structure **600** may include a lower surface of the body portion **621** of the second heat dissipation structure **620**. For example, the body portion **621** may be provided on the upper surfaces of the first semiconductor package **100**, the second semiconductor package **200**, and the first passive element **400**. The first heat conduction layer **710** may be in physical contact with the lower surface of the body portion **621** of the second heat dissipation structure **620**.

(61) The leg portion **622** of the second heat dissipation structure **620** may be provided between the edge region of the body portion **621** and the substrate **500**. The leg portion **622** of the second heat dissipation structure **620** may be connected to the body portion **621**. As shown in FIG. **2**A, the first semiconductor package **100**, the second semiconductor package **200**, and the first passive element **400** may be spaced apart from the leg portion **622** of the second heat dissipation structure **620**. The leg portion **622** may be provided in the edge region of the substrate **500** in plan view. The second heat dissipation structure **620** may include a thermally conductive material.

(62) The second heat dissipation structure **620** has electrical conductivity and may shield electromagnetic interference (EMI) of the first, second, and third semiconductor packages **100**, **200**, and **300**. The EMI means that electromagnetic waves that are radiated or conducted from an electrical element cause interference with the reception/transmission function of other electrical elements. By a second heat dissipation structure **620**, the operation of the first to third semiconductor packages **100**, **200**, **300** and the first passive element **400** may not interrupt the operation of the other package or may not be interrupted by the other package.

(63) Adhesive patterns **741** and **742** may be provided between the substrate **500** and the leg portions **622** of the second heat dissipation structure **620** to fix the second heat dissipation structure **620** to the substrate **500**. The adhesive patterns **741** and **742** may include a conductive adhesive pattern **741** and an insulating adhesive pattern **742**. The conductive adhesive pattern **741** may be provided between the ground pad **510**G and the leg portion **622** of the second heat dissipation structure **620**. The second heat dissipation structure **620** may be connected to the ground pad **510**G through the conductive adhesive pattern **741**.
(64) If more than a certain amount of charges is accumulated in the heat dissipation structure **600**, the charges may flow from the heat dissipation structure **600** into another electrically conductive component to damage the electrically conductive component. The electrically conductive component includes at least one of integrated circuits and wires in the first to third semiconductor chips **120**, **220** and **320**, wires in the first to third substrates **110**, **210** and **310**, the first to third connection terminals **150**, **250**, and **350**, and interconnections in the substrate **500**. According to example embodiments, a ground voltage may be applied to the second heat dissipation structure **620** by a conductive adhesive pattern **741**. Accordingly, the second heat dissipation structure **620** may limit and/or prevent electrical damage of the package system **1***b* due to electrostatic discharge (ESD).
(65) An insulating adhesive pattern **742** may be provided between the substrate **500** and the heat dissipation structure **600**. Accordingly, the heat dissipation structure **600** is insulated from the substrate **500**, so that the occurrence of electrical short may be limited and/or prevented. The thickness A**5** of the conductive adhesive pattern **741** may be substantially the same as the thickness of the insulating adhesive pattern **742**.
(66) The height B of the leg portion **622** of the second heat dissipation structure **620** may be less than the height H**1** of the mounted first semiconductor package **100**. At this time, the height B of the leg portion **622** may be equal to the height of the inner surface of the second heat dissipation structure **620**. The conductive adhesive pattern **741** may be in physical contact with the lower surface of the leg portion **622**. Accordingly, the thickness A**1** of the first heat conduction layer **710** may be less than the thickness of the adhesive patterns **741** and **742** (e.g., the thickness A**5** of the conductive adhesive pattern **741**). Since the thickness A**1** of the first heat conduction layer **710** is small, heat generated from the first semiconductor package **100** may be transferred to the heat dissipation structure **600** through the first heat conduction layer **710** more quickly.
(67) FIG. **2**C is a plan view showing a package system according to example embodiments. FIG. **2**D is a cross-sectional view taken along the line I-II of FIG. **2**C. Hereinafter, the contents overlapping with those described above will be omitted.
(68) Referring to FIGS. **2**C and **2**D, a package system **1***c* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The heat dissipation structure **600** may include the second heat dissipation structure **620** described with reference to FIGS. **2**A and **2**B. For example, the second heat dissipation structure **620** may include a body portion **621** and a leg portion **622**.
(69) A conductive adhesive pattern **741** may be provided between the ground pad **510**G and the leg portion **622** of the second heat dissipation structure **620** to connect with the second heat dissipation structure **620** and the ground pad **510**G. Unlike the example of FIGS. **2**A and **2**B, an insulating adhesive pattern **742** may not be provided. The thickness A**1** of the first heat conduction layer **710** may be less than the thickness A**5** of the conductive adhesive pattern **741**.
(70) The substrate **500**, the first to third semiconductor packages **100**, **200** and **300**, the first passive element **400** and the first to third heat conduction layers **710**, **720**, and **730** may be substantially the same as those described above with reference to FIGS. **1**A to **1**I.
(71) FIG. **2**E is a view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**C. Hereinafter, the contents

overlapping with those described above will be omitted.

(72) Referring to FIGS. **2**C and **2**E, a package system **1***d* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The substrate **500**, the first to third semiconductor packages **100**, **200** and **300**, the first passive element **400** and the first to third heat conduction layers **710**, **720**, and **730** may be substantially the same as those described above with reference to FIGS. **1**A to **1**E.

(73) The heat dissipation structure **600** may include a first heat dissipation structure **610**, a second heat dissipation structure **620**, and a heat dissipation layer **630**. The first heat dissipation structure **610** may be substantially the same as that described above with reference to FIGS. **1**A to **1**C. However, the first heat dissipation structure **610** may be disposed on the upper surface of the second heat dissipation structure **620**. The second heat dissipation structure **620** may be substantially the same as the second heat dissipation structure **620** described with reference to FIGS. **2**A to **2**D. For example, the second heat dissipation structure **620** may include a body portion **621** and a leg portion **622**. The width of the first heat dissipation structure **610** may be equal to or wider than the width of the second heat dissipation structure **620**. A conductive adhesive pattern **741** may be provided between the ground pad **510**G and the second heat dissipation structure **620**. As another example, an insulating adhesive pattern **742** as described in the example of FIGS. **2**A and **2**B may be further provided. The heat dissipation layer **630** may be interposed between the first heat dissipation structure **610** and the second heat dissipation structure **620**. The heat dissipation layer **630** may include, for example, a thermal interface material.

(74) FIG. **3**A is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**A. Hereinafter, the contents overlapping with those described above will be omitted.

(75) Referring to FIGS. **2**C and **3**A, a package system **1***e* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**.

(76) The first semiconductor package **100** includes a first adhesive layer **141** and a first thermal conductive structure **140** in addition to the first substrate **110**, the first semiconductor chip **120**, and the first molding layer **130**. The first thermal conductive structure **140** may have a relatively high thermal conductivity. The first thermal conductive structure **140** may include the thermally conductive material described in the example of FIGS. **1**A to **1**C. In one example, the first thermal conductive structure **140** may include a metal layer, a heat sink, or a heat pipe. As another example, the first thermal conductive structure **140** may use a water cooling method. The first adhesive layer **141** may be provided between the first molding layer **130** and the first thermal conductive structure **140**. The first adhesive layer **141** may include a thermal interface material. During an operation of the first semiconductor package **100**, heat generated from the first semiconductor chip **120** may be transferred to the first heat conduction layer **710** through the first adhesive layer **141** and the first thermal conductive structure **140**.

(77) According to example embodiments, the upper surface of the first semiconductor package **100** may correspond to the upper surface of the first thermal conductive structure **140**. The height H**1** of the mounted first semiconductor package **100** is defined as the sum of the height of the first connection terminals **150**, the height of the first substrate **110**, the height of the first molding layer **130**, the height of the first adhesive layer **141**, and the height of the first thermal conductive structure **140**. Even if the upper surface of the first molding layer **130** is provided at a lower level than the upper surface of the second semiconductor package **200** or the upper surface of the third semiconductor package **300**, by the provision of the first adhesive layer **141** and the first thermal conductive structure **140**, the height H**1** of the mounted first semiconductor package **100** may be greater than the height H**2** of the mounted second semiconductor package **200** and the height H**3** of the mounted third semiconductor package **300**. The thickness A**1** of the first heat conduction layer

710 may be less than the thickness A**2** of the second heat conduction layer **720** and the thickness A**3** of the third heat conduction layer **730**. Accordingly, the thermal characteristics of the first semiconductor package **100** may be improved.
(78) The substrate **500**, the second and third semiconductor packages **200** and **300**, the first passive element **400**, the first to third heat conduction layers **710**, **720**, and **730**, and the heat dissipation structure **600** may be substantially the same as those described with reference to FIGS. **1**A to **1**F and FIGS. **2**A to **2**E.
(79) FIG. **3**B is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**C. Hereinafter, the contents overlapping with those described above will be omitted.
(80) Referring to FIGS. **2**C and **3**B, a package system if includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The substrate **500**, the first semiconductor package **100**, and the first passive element **400**, the first to third heat conduction layers **710**, **720**, **730**, and the heat dissipation structure **600** may be substantially the same as those described above.
(81) The second semiconductor package **200** includes a second adhesive layer **241** and a second thermal conductive structure **240** in addition to the second substrate **210**, the second semiconductor chip **220**, and the second molding layer **230**. The second thermal conductive structure **240** may include a thermally conductive material and may have a relatively high thermal conductivity. The second thermal conductive structure **240** may include a metal layer, a heat sink, or a heat pipe. The second adhesive layer **241** may be provided between the second molding layer **230** and the second thermal conductive structure **240**. The second adhesive layer **241** may include a thermal interface material. During an operation of the second semiconductor package **200**, heat generated from the second semiconductor chip **220** may be transferred to the second heat conduction layer **720** through the second adhesive layer **241** and the second thermal conductive structure **240**.
(82) The upper surface of the second semiconductor package **200** may correspond to the upper surface of the second thermal conductive structure **240**. The height H**2** of the mounted second semiconductor package **200** is defined as the sum of the height of the second connection terminals **250**, the height of the second substrate **210**, the height of the second molding layer **230**, the height of the second adhesive layer **241**, and the height of the second thermal conductive structure **240**. The height H**1** of the mounted first semiconductor package **100** may be greater than the height H**2** of the mounted second semiconductor package **200**. Accordingly, the thickness A**1** of the first heat conduction layer **710** may be smaller than the thickness A**2** of the second heat conduction layer **720**.
(83) The third semiconductor package **300** includes a third adhesive layer **341** and a third thermal conductive structure **340** in addition to the third substrate **310**, the third semiconductor chip **320**, and the third molding layer **330**. The third thermal conductive structure **340** may include a thermally conductive material and may have a relatively high thermal conductivity. The third thermal conductive structure **340** may include a metal layer, a heat sink, or a heat pipe. The third adhesive layer **341** may be provided between the third molding layer **330** and the third thermal conductive structure **340**. The third adhesive layer **341** may include a thermal interface material. During an operation of the third semiconductor package **300**, heat generated from the third semiconductor chip **320** may be transferred to the third heat conduction layer **730** through the third adhesive layer **341** and the third thermal conductive structure **340**.
(84) The upper surface of the third semiconductor package **300** may correspond to the upper surface of the third thermal conductive structure **340**. The height H**3** of the mounted third semiconductor package **300** is defined as the sum of the height of the third connection terminals **350**, the height of the third substrate **310**, the height of the third molding layer **330**, the height of the third adhesive layer **341**, and the height of the third thermal conductive structure **340**. The

height H**1** of the mounted first semiconductor package **100** may be greater than the height H**3** of the mounted third semiconductor package **300**. Accordingly, the thickness A**1** of the first heat conduction layer **710** may be smaller than the thickness A**3** of the third heat conduction layer **730**.
(85) Unlike what is shown, the second adhesive layer **241** and the second thermal conductive structure **240** are omitted, and the second heat conduction layer **720** may be in direct contact with the upper surface of the second molding layer **230** as shown in FIG. **1**D. As another example, the third adhesive layer **341** and the third thermal conductive structure **340** are omitted, and the third heat conduction layer **730** may be in direct contact with the upper surface of the third molding layer **330**.
(86) FIG. **3**C is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**A. Hereinafter, the contents overlapping with those described above will be omitted.
(87) Referring to FIGS. **2**C and **3**C, a package system **1***g* includes a substrate **500**, a first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The substrate **500**, the first to third semiconductor packages **100**, **200** and **300**, the first passive element **400**, the first to third heat conduction layers **710**, **720**, and **730**, and the heat dissipation structure **600** are substantially the same as those described above.
(88) The first semiconductor package **100** may be substantially the same as that described in the example of FIG. **3**A. For example, the first semiconductor package **100** includes a first substrate **110**, a first semiconductor chip **120**, a first molding layer **130**, a first adhesive layer **141**, and a first thermal conductive structure **140**. The second semiconductor package **200** and the third semiconductor package **300** may be substantially the same as those described in the example of FIG. **3**B, respectively. The second semiconductor package **200** includes a second substrate **210**, a second semiconductor chip **220**, a second molding layer **230**, a second adhesive layer **241**, and a second thermal conductive structure **240**. The third semiconductor package **300** includes a third substrate **310**, a third semiconductor chip **320**, a third molding layer **330**, a third adhesive layer **341**, and a third thermal conductive structure **340**.
(89) The height H**1** of the mounted first semiconductor package **100** may be greater than the height H**2** of the mounted second semiconductor package **200** and the height H**3** of the mounted third semiconductor package **300**. The thickness A**1** of the first heat conduction layer **710** may be less than the thickness A**2** of the second heat conduction layer **720** and the thickness A**3** of the third heat conduction layer **730**.
(90) FIG. **4**A is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**C. FIG. **4**B is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**C. Hereinafter, the contents overlapping with those described above will be omitted.
(91) Referring to FIGS. **2**C, **4**A, and **4**B, a package system **1***h* or **1***i* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first heat conduction layer **710**, and a heat dissipation structure **600**. The substrate **500**, the first to third semiconductor packages **300**, the first passive element **400**, the first heat conduction layer **710**, and the heat dissipation structure **600** are substantially the same as those described above.
(92) As shown in FIG. **4**A, the package system **1***h* may not include the second heat conduction layer **720**. The sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710** may be greater than the height H**2** of the mounted second semiconductor package **200**.
(93) As shown in FIG. **4**B, the package system **1***i* may not include a third heat conduction layer **730**. The sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710** may be greater than the height H**3** of the mounted third

semiconductor package **300**.

(94) FIG. **4**C is a cross-sectional view showing a package system according to example embodiments, corresponding to a cross-section taken along the line I-II of FIG. **2**C. Hereinafter, the contents overlapping with those described above will be omitted.

(95) Referring to FIGS. **2**C and **4**C, a package system **1***j* includes a substrate **500**, first to third semiconductor packages **100**, **200** and **300**, a first passive element **400**, first to third heat conduction layers **710**, **720**, and **730**, and a heat dissipation structure **600**. The thickness A**1** of the first heat conduction layer **710** may be less than the thickness A**2** of the second heat conduction layer **720** and the thickness A**3** of the third heat conduction layer **730**.

(96) A fourth heat conduction layer **740** is provided between the first passive element **400** and the heat dissipation structure **600** so that it may be in physical contact with the upper surface of the first passive element **400** and the lower surface **600***b* of the heat dissipation structure **600**. The fourth heat conduction layer **740** may include a thermal interface material. Heat generated from the first passive element **400** may be transferred to the heat dissipation structure **600** through the fourth heat conduction layer **740**. The height H**1** of the mounted first semiconductor package **100** may be greater than the height H**4** of the mounted first passive element **400**. For example, the upper surface of the first semiconductor package **100** may be disposed at a higher level than the upper surface of the first passive element **400**. Accordingly, the thickness A**1** of the first heat conduction layer **710** may be smaller than the thickness A**4** of the fourth heat conduction layer **740**.

(97) As another example, either the second heat conduction layer **720** or the third heat conduction layer **730** may be omitted.

(98) In the description of FIGS. **3**A to **3**C and **4**A to **4**C, either the first heat dissipation structure **610** or the second heat dissipation structure **620** may be omitted. In this case, the heat dissipation layer **630** may not be provided.

(99) In the description of FIGS. **4**A to **4**C, the first semiconductor package **100** may further include a first adhesive layer **141** and a first thermal conductive structure **140**. The second semiconductor package **200** may further include a second adhesive layer **241** and a second thermal conductive structure **240**. The third semiconductor package **300** may further include a third adhesive layer **341** and a third thermal conductive structure **340**.

(100) FIG. **5**A is a cross-sectional view showing a semiconductor module according to example embodiments. FIG. **5**B is a view for explaining a second passive element according to example embodiments, and is a cross-sectional view showing an enlarged view of the region C of FIG. **5**A. FIG. **5**C is a view for explaining lower pads and conductive terminals according to example embodiments, and shows the enlarged region VI of FIG. **5**A. FIG. **5**D is a view for explaining lower pads according to example embodiments. Hereinafter, the contents overlapping with those described above will be omitted.

(101) Referring to FIGS. **1**A, **5**A, and **5**B, the semiconductor module **10** may include a board **1000** and a package system **1**. For example, a printed circuit board may be used as the board **1000**. Conductive pads **1500** may be provided on the upper surface **1000***a* of the board **1000**. The conductive pads **1500** may be electrically connected to internal wires (not shown) of the board **1000**. Electrical connection with the board **1000** in this specification may mean electrical connection with the internal wires of the board **1000**.

(102) The package system **1** described with reference to FIGS. **1**A to **1**C may be mounted on the board **1000** so that the semiconductor module **10** may be formed. As another example, the package system **1***a* of FIG. **1**F, the package system **1***b* of FIGS. **2**A and **2**B, the package system **1***c* of FIGS. **2**C and **2**D, the package system **1***d* of FIG. **2**E, the package system **1***e* of FIG. **3**A, the package system if of FIG. **3**B, the package system **1***g* of FIG. **3**C, the package system **1***h* of FIG. **4**A, the package system **1***i* of FIG. **4**B, or the package system **1***j* of FIG. **4**C is mounted on the board **1000**, so that the semiconductor module **10** may be formed. For convenience, the package system **1** of FIGS. **1**A to **1**C is shown and described with respect to the semiconductor module **10** mounted on

the board **1000**, but inventive concepts are not limited thereto.
(103) The packaging of the package system **1** includes providing the package system **1** on the board **1000** such that the conductive terminals **550** face the board **1000** and electrically connecting the conductive terminals **550** to the conductive pads **1500**. The pitch of the conductive terminals **550** may be substantially the same as the pitch P**4** of the conductive pads **1500**. The pitch P**4** of the conductive pads **1500** may be standardized. For example, the pitch P**4** of the conductive pads **1500** may satisfy the Joint Electron Device Engineering Council (JEDEC) standard. The pitch P**4** of the conductive pads **1500** may be large. For example, the pitch P**4** of the conductive pads **1500** may be 0.65 mm or more.
(104) When the first semiconductor package **100**, the second semiconductor package **200**, and the third semiconductor package **300** are directly mounted on the board **1000**, each of the pitch P**1** of the first connection terminals **150**, the pitch P**2** of the second connection terminals **250**, and the pitch P**3** of the third connection terminals **350** may be required to be substantially the same as the pitch P**4** of the conductive pads **1500**. According to example embodiments, the first semiconductor package **100**, the second semiconductor package **200**, and the third semiconductor package **300** may be connected to the board **1000** through the substrate **500**. Accordingly, the pitch P**1** of the first connection terminals **150**, the pitch P**2** of the second connection terminals **250**, and the pitch P**3** of the third connection terminals **350** are freely designed without being constrained by the pitch P**4** of the conductive pads **1500**.
(105) The pitch P**1** of the first connection terminals **150** may be smaller than the pitch P**4** of the conductive pads **1500**. For example, the pitch P**1** of the first connection terminals **150** may be 0.4 mm or less. Accordingly, the first connection terminals **150** are provided more densely, so that the planar area of the first semiconductor package **100** may be reduced. The pitch P**2** of the second connection terminals **250** and the pitch P**3** of the third connection terminals **350** may be smaller than the pitch P**4** of the conductive pads **1500**. For example, each of the pitch P**2** of the second connection terminals **250** and the pitch P**3** of the third connection terminals **350** may be 0.4 mm or less. Accordingly, the second semiconductor package **200** and the third semiconductor package **300** may be miniaturized. Since the first to third semiconductor packages **100**, **200**, **300** are miniaturized, the distances between the first to third semiconductor packages **100**, **200**, **300** may be reduced. Thus, the lengths of the electrical signal paths between the first to third semiconductor packages **100**, **200**, and **300** may be reduced. The operating speed and reliability of the package system **1** may be improved.
(106) The second passive element **420** may be mounted on the lower surface **1000***b* of the board **1000**. As shown in FIG. 5B, the second connection terminal portions **402** may be further provided between the board **1000** and the second passive element **420**. The second passive element **420** may be connected to the board **1000** through the second connection terminal portions **402**. The second connection terminal portions **402** may include, for example, a solder, a pillar, a bump, or a ball grid array. The height H**6** of the mounted second passive element **420** may be defined as including the height H**61** of the second connecting terminal portions **402**. For example, the height H**6** of the mounted second passive element **420** is equal to the sum of the height H**61** of the second connection terminal portions **402** and the height H**60** of the second passive element **420′** before being mounted. For example, the height H**6** of the mounted second passive element **420** may be greater than the sum of the height H**1** of the mounted first semiconductor package **100** and the thickness A**1** of the first heat conduction layer **710**. Even if the height H**6** of the mounted second passive element **420** is large, the second passive element **420** may be electrically connected to the package system **1** through the substrate **500**.
(107) The second passive element **420** may be electrically connected to one of the first to third semiconductor packages **100**, **200**, and **300**. The second passive element **420** may be provided overlapping or adjacent to the one of the semiconductor packages **100**, **200**, and **300** in plan view. Accordingly, the length of the electrical signal path between the second passive element **420** and

the one of the semiconductor packages **100**, **200**, and **300** may be reduced. Thus, the electrical characteristics of the semiconductor module **10** may be improved.
(108) The second passive element **420** may be provided in plurality. In this case, the heights H**6** of the second passive elements **420** may be equal to or different from each other. The number of the second passive elements **420** may be variously modified. Hereinafter, referring to FIGS. **5**C and **5**D, the conductive terminals **550** and the lower pads **540** will be described.
(109) The lower pads **540** may be provided on the lower surface of the substrate **500**. The lower pads **540** may include a connection pad **541** and a test pad **542**. During the manufacturing process of the package system **1** or before the package system **1** is mounted on the board **1000**, the electrical characteristics of the package system **1** may be evaluated. Evaluation of the electrical characteristics may be performed using the test pad **542**. For example, as a probe (not shown) contacts the test pad **542**, the electrical characteristics and the connection relationship of at least one of the first to third semiconductor packages **100**, **200**, and **300**, the first passive element **400**, and the electronic element **430** may be evaluated. Thereafter, the conductive terminals **550** are formed, and the package system **1** may be mounted on the board **1000**.
(110) As shown in FIG. **5**C, the conductive terminals **550** may include a first terminal **551** and a second terminal **552**. The first terminal **551** is provided on the lower surface of the connection pad **541** and may be connected to the connection pad **541** and corresponding one of the conductive pads **1500**. The first terminal **551** may electrically connect the package system **1** to the board **1000**. The first terminal **551** may function as a path for signal transmission.
(111) The second terminal **552** is provided on the lower surface of the test pad **542** and may be connected to the test pad **542**. For example, the second terminal **552** may function as a ground terminal. The ground voltage is transmitted to the package system **1** through the board **1000** and the second terminal **552**. As another example, the second terminal **552** may be a dummy terminal. For example, the second terminal **552** may not be electrically connected to an internal wire in the board **1000**. Alternatively, the second terminal **552** may not be electrically connected to the package system **1**.
(112) As shown in FIG. **5**D, the second terminal (**552** in FIG. **5**C) may not be provided. The test pad **542** may be spaced apart from the board **1000** and electrically insulated. Although not shown in the drawing, the underfill material may be filled in the gap between the board **1000** and the test pad **542**. The underfill material may include an insulating polymer.
(113) According to inventive concepts, during an operation of the package system, the first semiconductor package may generate a lot of heat. The thickness of the first heat conduction layer may be less than the thickness of the second heat conduction layer and the thickness of the third heat conduction layer. As the thickness of the first heat conduction layer decreases, the thermal characteristics of the first semiconductor package may be improved. The package system may show improved operating characteristics.
(114) Although some example embodiments of inventive concepts have been described, it is understood that inventive concepts should not be limited to these embodiments but various changes and modifications may be made by one ordinary skilled in the art within the spirit and scope of inventive concepts as hereinafter claimed.

Claims

1. A semiconductor package system comprising: a substrate; a first semiconductor package on the substrate; a second semiconductor package laterally spaced apart from the first semiconductor package on the substrate; a first passive element on the substrate, the first passive element being spaced apart from the first semiconductor package; dam structures disposed between the second semiconductor package and the first passive element, and provided on the substrate; a heat dissipation structure on the first semiconductor package, the second semiconductor package, and

the first passive element; and a first heat conduction layer between the first semiconductor package and the heat dissipation structure, a second heat conduction layer between the second semiconductor package and the heat dissipation structure, wherein a sum of a first height of the first semiconductor package and a first thickness of the first heat conduction layer is greater than a height of the first passive element, wherein the first height of the first semiconductor package is greater than a second height of the second semiconductor package, wherein the first heat conduction layer and the second heat conduction layer include a thermal interface material comprising polymers and thermally conductive particles, and wherein the sum of the height of the first semiconductor package and the first heat conduction layer is the same as the sum of the height of the second semiconductor package and the second heat conduction layer.

2. The semiconductor package system of claim 1, further comprising: an underfill layer provided in a gap between the substrate and the second semiconductor package, wherein the dam structures are disposed between the underfill layer and the first passive element, and wherein the underfill layer is spaced apart from the first passive element.

3. The semiconductor package system of claim 2, wherein at least one dam structure of the dam structures is in physical contact with the underfill layer.

4. The semiconductor package system of claim 2, further comprising: solder terminals between the substrate and the second semiconductor package, wherein the underfill layer seals the solder terminals.

5. The semiconductor package system of claim 1, wherein viewed in plan, the dam structures surround the second semiconductor package and are spaced apart from the second semiconductor package.

6. The semiconductor package system of claim 1, wherein a height of the dam structures is equal to or less than the sum of the first height of the first semiconductor package and the first thickness of the first heat conduction layer.

7. The semiconductor package system of claim 1, further comprising: wherein the first thickness is less than a second thickness of the second heat conduction layer, and wherein the first heat conduction layer and the second heat conduction layer are in physical contact with the heat dissipation structure, and wherein the first heat conduction layer directly contacts an upper surface of the first semiconductor package.

8. The semiconductor package system of claim 1, further comprising: a ground pattern on an upper surface of the substrate; and a conductive adhesive pattern between the ground pattern and the heat dissipation structure, wherein the heat dissipation structure includes: a body portion extending in parallel to the upper surface of the substrate; and a leg portion between the conductive adhesive pattern and the body portion, the leg portion connected to the body portion, wherein the heat dissipation structure is electrically connected to the ground pattern through the conductive adhesive pattern, and wherein the first thickness is less than a thickness of the conductive adhesive pattern.

9. The semiconductor package system of claim 1, wherein the first semiconductor package includes a first substrate, a first semiconductor chip, and a first molding layer, wherein the first semiconductor chip is a system-on-chip, wherein the second semiconductor package includes a second substrate, a second semiconductor chip, and a second molding layer, and wherein the second semiconductor chip is a power management chip.

10. The semiconductor package system of claim 1, wherein the first semiconductor package includes a first semiconductor chip including one or more logic circuits, the first semiconductor package and the second semiconductor package are on an upper surface of the substrate, the first passive element is on the upper surface of the substrate, the dam structures disposed between the second semiconductor package and the first passive element are on the upper surface of the substrate; the first heat conduction layer and the second heat conduction layer are among a plurality of heat conduction layers that are each in physical contact with a lower surface of the heat dissipation structure, the first heat conduction layer is on an upper surface of the first

semiconductor package, and the first heat conduction layer has a thinnest thickness among the plurality of heat conduction layers.
11. The semiconductor package system of claim 10, further comprising: an underfill layer provided in a gap between the substrate and the second semiconductor package, wherein the underfill layer is spaced apart from the first passive element by the dam structures.
12. The semiconductor package system of claim 10, wherein the dam structures are spaced apart from each other.
13. The semiconductor package system of claim 10, wherein the dam structures include a same material as a material of an uppermost layer of the substrate, and wherein the dam structures are connected to the uppermost layer of the substrate.
14. The semiconductor package system of claim 10, wherein the dam structures include an insulating polymer.
15. The semiconductor package system of claim 10, wherein the second semiconductor package includes a package substrate and a second semiconductor chip, and the second semiconductor chip includes a power management integrated circuit.
16. The semiconductor package system of claim 10, further comprising: an underfill layer provided in a gap between the substrate and the second semiconductor package; solder terminals between the substrate and the second semiconductor package, a board on a lower surface of the substrate; conductive terminals connected to the substrate and the board; and a second passive element on a lower surface of the board, wherein the dam structures are disposed between the underfill layer and the first passive element, wherein the underfill layer is spaced apart from the first passive element, wherein the underfill layer seals the solder terminals, wherein a height of the second passive element is greater than the first height of the first semiconductor package, and wherein a pitch of the solder terminals is smaller than a pitch of the conductive terminals.
17. A semiconductor package system comprising: a substrate; a first semiconductor package on the substrate; a second semiconductor package laterally spaced apart from the first semiconductor package on the substrate; a first passive element on the substrate, the first passive element being spaced apart from the first semiconductor package; dam structures disposed between the second semiconductor package and the first passive element, and provided on the substrate; a heat dissipation structure on the first semiconductor package, the second semiconductor package, and the first passive element; a first heat conduction layer between the first semiconductor package and the heat dissipation structure; an underfill layer provided in a gap between the substrate and the second semiconductor package; solder terminals between the substrate and the second semiconductor package, a board on a lower surface of the substrate; conductive terminals connected to the substrate and the board; and a second passive element on a lower surface of the board, wherein a sum of a first height of the first semiconductor package and a first thickness of the first heat conduction layer is greater than a height of the first passive element, wherein the first height of the first semiconductor package is greater than a second height of the second semiconductor package, wherein the dam structures are disposed between the underfill layer and the first passive element, wherein the underfill layer is spaced apart from the first passive element, wherein the underfill layer seals the solder terminals, wherein a height of the second passive element is greater than the first height of the first semiconductor package, and wherein a pitch of the solder terminals is smaller than a pitch of the conductive terminals.
18. A semiconductor package system comprising: a first substrate; a first semiconductor package on the first substrate, the first semiconductor package including a first semiconductor chip, the first semiconductor chip including one or more logic circuits; a second semiconductor package on the first substrate; an underfill layer provided in a gap between the first substrate and the second semiconductor package, a passive element on the first substrate; dam structures disposed between the underfill layer and the passive element on the first substrate; a heat dissipation structure on the first semiconductor package, the second semiconductor package, and the passive element; a first

heat conduction layer on the first semiconductor package, the first heat conduction layer in physical contact with the heat dissipation structure; and a second heat conduction layer on the second semiconductor package, the second heat conduction layer in physical contact with the heat dissipation structure, wherein a thickness of the first heat conduction layer is smaller than a thickness of the second heat conduction layer, wherein an upper surface of the first heat conduction layer is provided at a level higher than an upper surface of the passive element, wherein the upper surface of the first heat conduction layer is at a substantially same level with an upper surface of the second heat conduction layer, wherein the second semiconductor package includes a second substrate, a second semiconductor chip, and a molding layer, wherein the first substrate is spaced apart from the second substrate, wherein the upper surface of the first heat conduction layer is at a substantially same level with a lower surface of the heat dissipation structure, and wherein the second semiconductor chip is a different type of semiconductor chip than the first semiconductor chip, wherein the first heat conduction layer is in physical contact with the first semiconductor package, and wherein a lower region of the first heat conduction layer and an upper region of the first heat conduction layer comprise a same material composition.

19. The semiconductor package system of claim 18, wherein the underfill layer is spaced apart from the passive element by the dam structures, and wherein the dam structures are vertically spaced apart from the lower surface of the heat dissipation structure.

20. The semiconductor package system of claim 18, further comprising: a third semiconductor package on the first substrate; and a third heat conduction layer on the third semiconductor package, wherein the third heat conduction layer is in physical contact with the heat dissipation structure, wherein the thickness of the first heat conduction layer is smaller than a thickness of the third heat conduction layer, wherein the first heat conduction layer and the second heat conduction layer include a thermal interface material comprising polymers and thermally conductive particles, and wherein a sum of a height of the first semiconductor package and a first thickness of the first heat conduction layer is the same as a sum of a height of the second semiconductor package and a second thickness of the second heat conduction layer.